Autonomous Underwater Vehicle

Grippers for Underwater Manipulation

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Abstract

Underwater manipulation remains one of the most challenging unsolved problems in marine robotics. Existing industrial grippers are mostly rigid, which means they can crush delicate specimens like corals. Soft grippers are gentler but tend to tear under repeated pressure cycling at depth. Our AUV that we worked on earlier in ILGC could already navigate and detect objects, but had no reliable way to physically interact with them. Therefore, this project set out to design, prototype, and test a general-purpose underwater gripper capable of grasping a range of objects, including spherical, thin, slippery, and rigid ones.

We approached this by adopting the fin-ray effect, a mechanism inspired by the bone structure of ray-finned fish, where flexible fingers passively wrap around an object on contact instead of squeezing it from one direction. Starting from an open-source 3-finger design, we moved to a 4-finger configuration, redesigned the worm-follower actuation base, changed the internal structure of the finger arms, and moved the pivot point outward to increase the opening range. Parts were 3D printed in SLA, TPU and ABS.

However, trial and error revealed a clear trade-off: moving the pivot outward improved how wide the gripper could open, but the fingertips no longer met at the centre during closure. This left a gap that stopped the gripper from grasping small or thin objects. Two fixes were explored: adding a curve to the pivot region of the finger arm, and changing the slit angles in the worm-follower base. In parallel, we investigated a rack-and-pinion mechanism where a central gear drives all fingers at once, completely removing the dependency on pivot placement. The rack-and-pinion prototyping introduced its own problems around lateral drift of the finger and structural fragility, which were resolved through revised clamp geometry and increased wall thickness. The testing plan has been fully defined and will be executed before the end-semester presentation.

Future work includes silicone moulding of finger surfaces for better grip on slippery objects. This project lays the foundation for a capable, modular underwater gripper that can be handed over to the next cohort for completion and field testing.

1. Introduction

1.1 Motivation

In the previous semester of ILGC, our team worked on an Autonomous Underwater Vehicle capable of navigating underwater and identifying objects in its environment. While the vehicle could move toward and locate objects, it lacked a reliable way to physically interact with them. This gap between detection and manipulation was what motivated the gripper project. With a dependable gripper, the AUV could collect samples, retrieve objects, and perform meaningful tasks underwater rather than just observing them.

This problem turns out to be harder than it first appears. Gripping objects underwater is fundamentally different from gripping them in air. As a gripper's fingers approach an object, the surrounding water creates drag that pushes the object away before contact is even made. Once contact happens, wet surfaces are far more slippery, making it harder to maintain a secure hold. Buoyancy also changes the effective weight of the object, which affects how much force is needed to lift or move it. Together, these factors make underwater grasping significantly more challenging than the same task in air (Mazzeo et al., 2022a)

The gripper designs used in industrial underwater robotics today are mostly rigid, using parallel fingers or claw-style mechanisms. These work well for hard, robust objects but are poorly suited to anything fragile. Corals, sponges, and biological specimens can be crushed or damaged by the concentrated contact forces exerted by rigid fingers(Mazzeo et al., 2022b). Research has been moving toward softer gripper designs that wrap around an object more gently, distributing force over a larger contact area. Soft grippers are more tolerant of small positioning errors and less likely to damage delicate specimens. However, soft materials have lower mechanical strength and tend to tear under the repeated pressure changes that occur when a robot dives and resurfaces (Mazzeo et al., 2022a). Our goal was to find a design that sits between these two extremes, robust enough to function reliably but gentle enough to handle fragile objects without causing damage.

1.2 Problem Statement

The goal of this project was to design, prototype, and iteratively improve an underwater robotic gripper for the AUV. The gripper needed to reliably grasp a range of object types, including thin, circular, delicate, slippery, and rigid objects. It also needed to meet practical constraints: it had to be lightweight to avoid adding too much to the vehicle, water-resistant to survive the operating environment, and simple enough to be actuated by a single drive motor.

2. Literature Review

Given the physical challenges described above, designing an effective underwater gripper requires more than engineering intuition, it demands a systematic understanding of what such a gripper actually needs to do, what currently exists, and where existing solutions fall short. The following review builds that understanding in three steps: first, establishing what manipulation tasks and requirements define the problem space, then examining the landscape of existing gripper designs and their limitations, and finally reviewing the mechanical principles that inform a more capable approach.

2.1 Taxonomy of Underwater Manipulative Actions

Before evaluating any gripper design, it is necessary to understand what underwater manipulation actually involves at a task level. A foundational contribution here was made by Mazzeo et al. (2022a), who developed the first formal taxonomy of underwater manipulative actions for deep-sea specimen collection. Their methodology involved structured interviews with seventeen domain experts, marine biologists, ROV pilots, and geologists, supplemented by analysis of field video documentation. The study systematically categorised underwater manipulation into three primary task types: detachment, collection, and storage. Detachment actions ranged from non-retaining forms, such as pushing and scraping, to prehensile retaining forms, including grip-and-twist and cage-and-pull. The collection encompassed gripping, caging, suction, and tool-gripping. Storage included suction-based transfer, pouring, and object release. The taxonomy further organises these into four hierarchical levels: the full manipulation sequence, the atomic manipulation (a specific object interaction), the key-event (the period between contact creation and release), and the manipulation primitive (a basic robot command such as move or exert force). Manipulation goals are additionally classified into hand-only actions, separation actions, and release-determined actions. This decomposition is useful for mapping gripper requirements to specific task stages rather than treating manipulation as a single undifferentiated action.

Each task type maps to specific gripper contact configurations: point, edge, flat, hook, hollow, grasp, envelope, and suction flow. This framework provided a principled basis for evaluating whether any given gripper design is mechanically fit for the range of manipulation tasks encountered underwater.

The study also identified the primary failure modes of current underwater manipulators: squashing of soft specimens on full closure, poor force control dependent on pilot experience, slow closure speed that allows mobile organisms to escape, absence of force and tactile feedback, and impractical tool-change procedures in currents. These failure modes directly motivated the transition in this project from the prior cup-with-lid manipulator to a compliant, force-distributing gripper design.

Engineering requirements synthesised from this taxonomy span three dimensions:

1. **Operational:** grippers must be lightweight, energy-efficient, corrosion-resistant, and compatible with the arm's actuation and communication systems.
2. **Environmental:** they must be fully water and pressure-resistant, near-neutrally buoyant, and functional across ocean temperatures of 0°C to 3°C.
3. **Task-related:** requirements include high-resolution position and force control, reactive closure speed, and the capacity to handle objects ranging from sub-centimetre specimens up to 15 cm (Mazzeo et al., 2022a).

2.2 Existing Gripper Mechanisms and Trends

With a clear picture of what underwater grippers need to accomplish, the next question is whether existing designs meet those requirements. A companion systematic review by Mazzeo et al. (2022b) surveys the state of existing underwater grippers and organises them into four mechanical classes: parallel fingers, opposed or intermeshed fingers, grabber claws, and cage claws. Actuation is primarily hydraulic for deep-sea ROV-deployed systems, electric for laboratory and pool prototypes, and occasionally pneumatic. Associated storage systems include suction samplers, scoops, sediment corers, and Niskin bottles.

The review distinguishes prototype maturity by testing environment. Lab-tank prototypes typically combine rigid and soft technology with electric, hydraulic, or pneumatic actuation. Prototypes tested in pools or open sea conditions are predominantly purely rigid or purely soft with electric actuation. Grippers deployed on deep-sea ROVs tend toward fewer joints and actuators and are mainly hydraulically actuated. This gradient reveals a clear pattern: design complexity and softness decrease as deployment depth and operational demands increase, reflecting the current limits of soft robotic durability in high-pressure, real-world conditions.

The review identifies a consistent gap: research prototypes tend to lack robustness outside their designed scenarios, while operationally deployed marine grippers closely resemble rigid industrial designs offering limited adaptability to irregular specimen shapes. Tactile sensing remains an unsolved problem, with current workarounds relying on mechanical stiffness selection through spring-based compliance, adjusted transmission, or jamming structures, rather than direct contact feedback.

A significant documented trend is therefore a shift toward bio-inspired, soft gripper designs that address these gaps. Soft grippers offer tolerance to positioning errors and reduced risk of damage to fragile specimens. Their limitations include lower mechanical strength and susceptibility to tearing under cyclic pressurisation, a critical consideration for repeated operation. Ongoing mitigation strategies include fibre reinforcement, improved interfaces between soft and rigid components, and modular construction for simplified maintenance (Mazzeo et al., 2022b). This sets up the central design question for this project: what mechanical principles make a soft gripper both compliant and reliable?

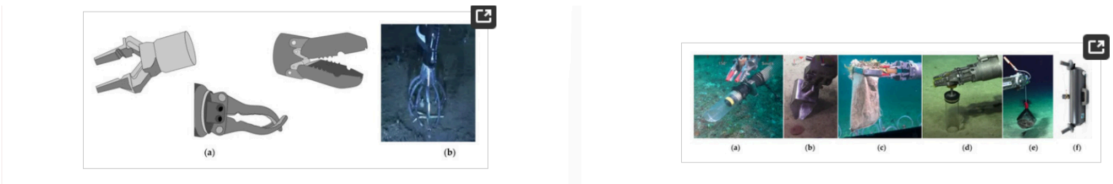

Fig1. Various types of Gripper and Collection tools highlighted in the paper

(Mazzeo, A., Aguzzi, J., Calisti, M., Canese, S., Vecchi, F., Stefanni, S., & Controzzi, M. (2022). Marine robotics for deep-sea specimen collection: A systematic review of underwater grippers. Sensors, 22(2), Article 648. https://doi.org/10.3390/s22020648)

2.3 Grasping Forces: Normal and Shear Mechanisms

Understanding why soft grippers perform better on fragile objects requires a closer look at the mechanics of grip itself. Hawkes et al. (2015) provide a useful framework, distinguishing between normal-force and shear-force grasping strategies. Normal-force approaches include wrapped or form-closure grasps, where the gripper geometrically surrounds the object and friction is not required, and squeeze or force-closure grasps, where inward clamping generates sufficient friction for a stable hold. Shear-force approaches instead exploit surface microstructure: gecko-inspired silicone ridges that activate Van der Waals adhesion forces when loaded laterally, increasing contact area without requiring compressive clamping.

This distinction is directly relevant for underwater design. The aquatic environment reduces surface friction on wet and smooth objects, making pure force-closure strategies unreliable. Form-closure, by contrast, does not depend on friction at all, making it inherently better suited to the underwater context. The fin-ray gripper pursued in this project primarily employs a form-closure strategy, where compliant fingers conform to the object's geometry and distribute load across a large contact area. The planned addition of silicone surface features draws on the Hawkes et al. (2015) finding that shear-based adhesion can supplement conformational wrapping in scenarios where geometric enclosure alone is insufficient.

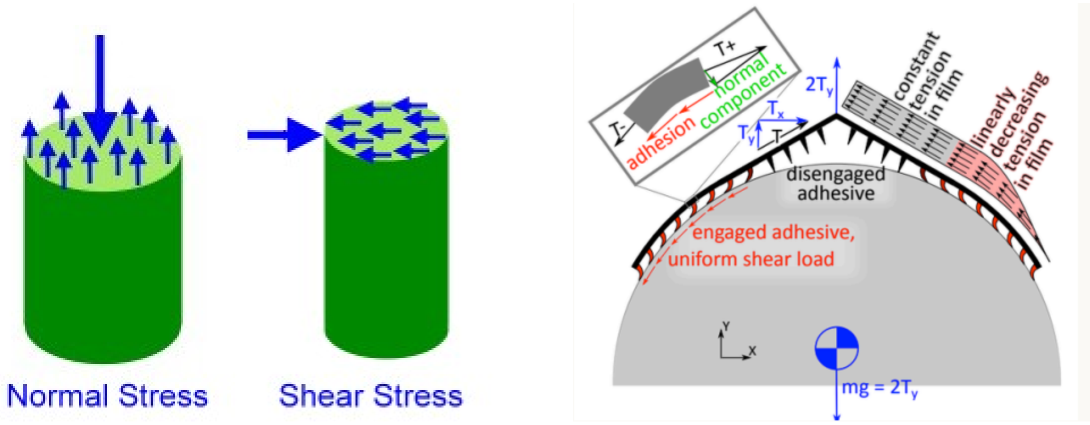

Fig2. Gripping using Shear Forces

(Hawkes, E.W., Christensen, D.L., Han, A.K., Jiang, H. and Cutkosky, M.R., 2015. Grasping without squeezing: Shear adhesion gripper with fibrillar thin film. In 2015 IEEE ICRA (pp. 2305-2312))

2.4 Fin-Ray Effect and Structural Compliance

Form-closure grasping requires fingers that can passively conform to an object's shape without active sensing or control. The Fin Ray Effect, derived from the mechanical behaviour of fish pectoral fins, provides exactly this. The geometry consists of two flexible outer beams connected by internal transverse ribs, forming a series of compliant triangular cells. When a lateral load is applied to the tip, rather than deflecting away from the force, the structure bends toward it. This counterintuitive response enables passive mechanical compliance: the finger conforms to an object's contour, increasing contact area while reducing localised pressure.

Srinivas et al. (2024) systematically investigated how different internal filling geometries affect the performance of 3D-printed fin-ray fingers, testing multiple rib and infill patterns against objects of varying shape and stiffness, including coffee cups and wooden blocks. Their key finding was that internal structure significantly influences the bending profile and grip strength of the finger, with different infill patterns suited to different object types. This provides the direct rationale for prototyping and comparing multiple internal finger geometries in this project, including the original transverse-rib design and a triangulated infill variant. The study also confirms TPU (thermoplastic polyurethane) as an appropriate finger material given its elastic memory, resistance to cyclic deformation, and printability (Srinivas et al., 2024).

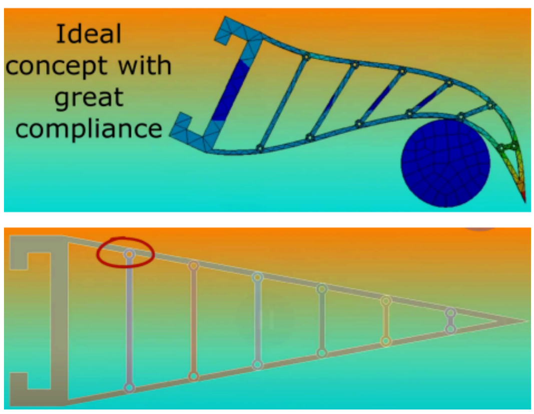

Fig. 3: Finray gripper deforming to envelop the object

(Image from RoboSub 2025 Technical Design Report Nanyang Technological University (NTU))

Together, these studies form the foundation for the iterative design and prototyping approach described in the following sections.

3. Modelling & Analysis - Design Modifications

This section documents the full design evolution of the underwater gripper mechanism. We started with the worm-follower fin-ray gripper and then iterated it from a 3-finger to a 4-finger design. After encountering issues with the worm follower geometry, we then shifted to rack-and-pinion architecture, culminating in the decision to fabricate the final design in aluminium for structural durability.

3.1 Worm-Follower Fin-Ray Gripper

3.1.1 Design Concept and Initial Configuration

The first gripper mechanism designed for this project was based on a worm-follower actuation system combined with fin-ray fingers. In this mechanism, a motor-driven worm gear drives a follower that translates rotational motor output into the radial motion of the finger arms. All components were modelled in Fusion 360 and 3D printed in ABS and TPU. The initial printed assembly comprised a worm-follower base, a front base plate, fin-ray fingers (TPU material), and a motor-mount holder. The original design began as a 3-finger configuration, adapted from reference designs and modified in-house to suit the grasping requirements of the project.

3.1.2 3-Finger to 4-Finger Design Modification

The original 3-finger gripper (Fig. 4 (a)) was modified to a 4-finger(Fig. 4(b)) configuration as an iterative improvement. The rationale for this change was to test whether an additional finger arm would improve grasping stability, distribute grip force more evenly, and increase success rates when grasping larger, rounder, or asymmetric objects.

The following changes were made to the worm-follower base to support four fingers:

a. The base was redesigned to accommodate four equidistant arm channels instead of three.
b. The slit geometry was enlarged to allow greater freedom of movement for the rivets acting as pivot fasteners.
c. Hooks were added to the base to pull the finger arms upright after each grasping cycle, ensuring consistent return to the open position.

Fig. 4 (a) 3-finger configuration

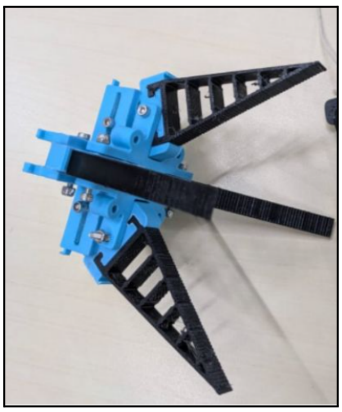

Fig. 4 (b) 4-finger configuration

3.1.3 Pivot Relocation and the Closure Trade-Off

To improve the opening range of the gripper, enabling it to grasp larger and rounder objects, the pivot location of the finger arms was shifted outward(Fig. 5). This modification confirmed improved range of motion.

However, upon actuation, a significant and unexpected problem emerged: the fingers were not closing fully. Moving the pivot outward shifted the effective closing axis of each finger arm, meaning the fingertips no longer converged at the centre during closure. The result was a persistent gap at the fingertips that limited the gripper's ability to grasp small and thin objects, a direct trade-off with the improved opening range.

Two corrective approaches were evaluated to restore full closure while retaining the wider opening range:

a. Adding a curve to the pivot region of the finger arm to redirect the closing trajectory back toward the centre, compensating geometrically for the axis shift.
b. Modifying the slit angles in the worm-follower base to alter the rivet travel path and correct the closing axis without changing the finger arms themselves.

Neither approach fully resolved the trade-off. The geometric constraints of the worm-follower mechanism meant that pivot location, opening range, and closure geometry were inherently coupled and improving one parameter degraded another. This limitation made us explore an alternative actuation mechanism that could decouple these constraints entirely

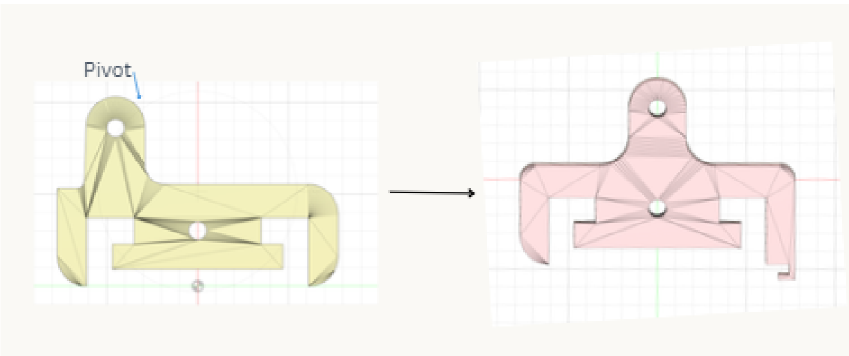

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Fig. 5: Change in pivot point of Clamp

3.2 Transition to the Rack-and-Pinion Design

3.2.1 Rationale for the Mechanism Change

The rack-and-pinion gripper(Fig. 6) was selected as the replacement architecture after a systematic analysis of the worm-follower's structural limitations. The rack-and-pinion mechanism resolves this at the kinematic level:

- A central pinion gear, driven by the motor, meshes directly with rack teeth cut into the base of each finger.
- As the pinion rotates, all fingers are driven simultaneously and symmetrically - inward to close, outward to open.

- Both full closure and maximum opening are governed purely by rack travel length and pinion rotation, completely independent of any pivot geometry.

This architecture is also compact, straightforward to model in Fusion 360, and ensures perfect synchrony across all four fingers without additional linkage or geometry changes. We adapted an existing open-source 3-finger rack-and-pinion design (PrintChallenge) and re-engineered it for a four-finger.

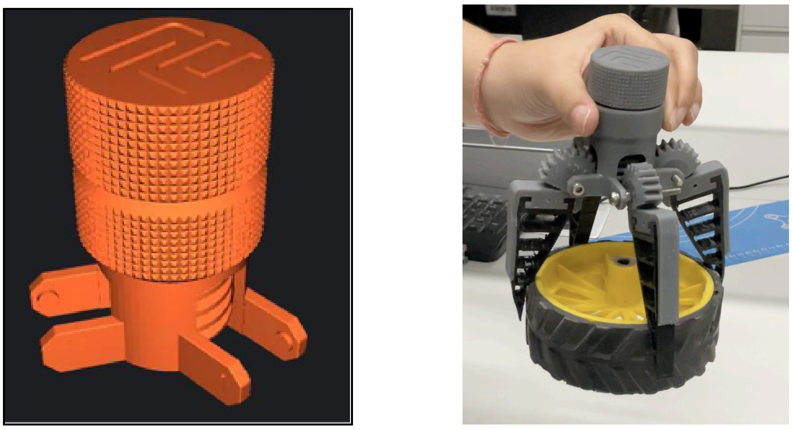

Fig. 6: Rack and pinion design

3.2.2 Rack-and-Pinion Design Iterations

Following the mechanism transition, the rack-and-pinion gripper underwent four distinct design iterations. Each iteration was driven by a specific failure observed during physical prototyping.

Iteration 1 - Initial Clamp: Lateral Drift Failure

Upon assembling and actuating the initial clamp design, the finger arms were found to deflect tangentially, sliding sideways perpendicular to their intended radial axis of motion. This lateral drift caused misalignment during closure, preventing a clean and symmetric grip.

Root cause: No lateral constraint geometry in the clamp channels; the finger arms had freedom to move tangentially as well as radially.

Corrective action: The clamp was redesigned to incorporate lateral guide walls that confine each finger arm strictly to its radial axis, eliminating tangential drift.

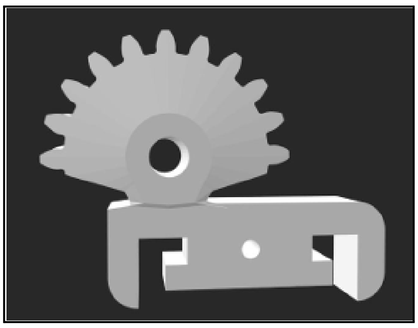

Fig 7(a): Original clamp for Rack and pinion.

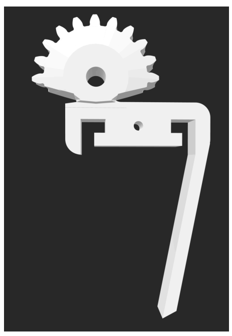

Fig. 7(b): Clamp with support for Rack and pinion

Iteration 2 - Pin-and-Ray Mechanism: Structural Failures

During printing and assembly of the pin-and-ray mechanism, two structural failures were observed:

a. Base fracture: The bottom section was thin, causing it to fracture under assembly loads.
b. Side wall breakage: The walls between the two finger clamp holders lacked bridging support and were separating under actuation.

Corrective actions: Base thickness was increased. Reinforcing support geometry was added between the clamp holders to prevent separation.

Fig8(a): Original Rack and Pinion

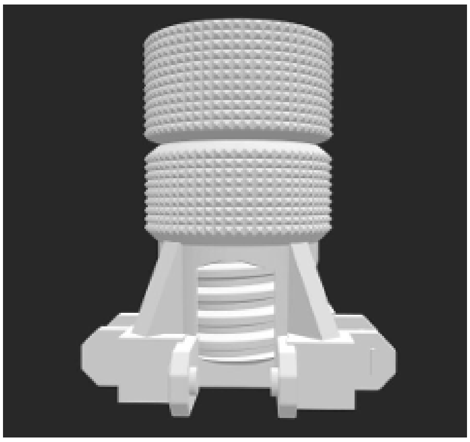

Fig8(b): Redesigned Rack and pinion

Iteration 3 - SLA Print Constraint: Splitting the Assembly

After finalising the revised geometry, a manufacturing constraint was identified: the screw-and-clamp-holder assembly could not be printed as a single part on the SLA printer. SLA printing was required because ABS was weak and produced an insufficient surface finish on the screw interface. Rough surfaces introduced excessive friction, degrading the mechanism's performance. SLA's finer resolution and smoother surfaces were necessary for the tight-tolerance screw travel.

Solution: The assembly was split into two separately printable parts:

1. the screw section and
2. the clamp holder section

to be joined post-print.

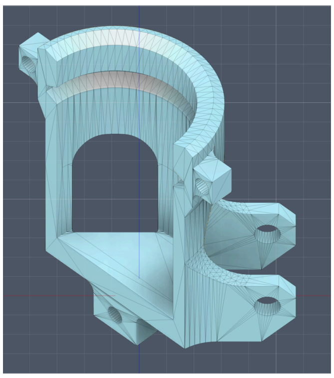

Fig. 9 Split Clamp Holder

Iteration 4 - SLA Material Failure: Transition to Aluminium

Despite resolving the print geometry, SLA-printed parts fractured under operational loads. When mechanical pressure was applied during screw actuation, the structure cracked. Repeated loading cycles during grasping exceeded the fatigue tolerance of the material.

Solution: We decided to fabricate the rack-and-pinion body in aluminium. Aluminium provides the surface finish quality required for smooth screw motion while delivering the structural strength and fatigue resistance that neither ABS nor SLA printing could offer simultaneously. The aluminium build is currently underway.

3.3 Fin-Ray Finger Internal Fill Geometry Variations

A key design modification carried out in parallel with the actuation mechanism work was the modification of the internal fill geometry of the fin-ray fingers. The standard fin-ray finger relies on its internal lattice structure to provide passive compliance, i.e., the finger deflects and wraps around an object on contact. The specific geometry of that internal structure directly governs the stiffness, compliance, and contact behaviour of the finger, making it a critical design variable for grasping different object types.

We designed and printed two distinct internal fill variants to evaluate how geometry affects grasping suitability:

Variant 1 - Diagonal Cross-Braced Fill

The first variant features a diagonal cross-braced internal structure(Fig. 9). The fill consists of angled struts forming irregular triangular and trapezoidal cells that run diagonally across the width of the finger. This geometry creates stiffer, more directionally rigid fingers. The diagonal bracing resists deformation along the loading axis more effectively than orthogonal patterns.

The diagonal cross-bracing also results in a sharper, more pointed contact profile at the finger tip, making it well-suited to grasping square-edged and sharp-cornered objects. The strut geometry allows the finger to lock around angular features rather than slipping off curved or smooth surfaces.

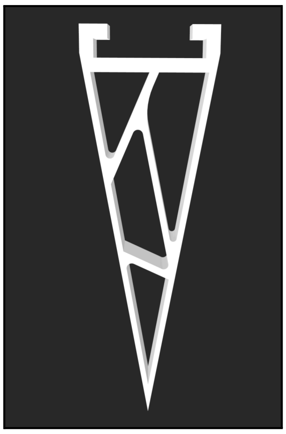

Fig 9: Diagonal Cross-Braced Fill

Variant 2 - Horizontal Ladder Fill with Microwedges

The second variant uses a horizontal ladder-style internal fill - a series of evenly spaced transverse ribs spanning the full width of the finger, creating uniform rectangular cells along the length of the finger(Fig. 10(a)). The outer edges of this variant also incorporate a microwedge profile along one side of the finger body to increase friction and grip strength on contact surfaces underwater.

The ladder fill distributes compliance more evenly along the length of the finger, allowing the finger to flex gradually and conform to curved or irregular surfaces. The uniform rib spacing ensures consistent stiffness across the finger length without concentrating stress at any single point. The microwedges serve a dual purpose: they increase the surface area in contact with an object and introduce mechanical interlocking features that improve grip friction on smooth, rounded, or slippery surfaces.

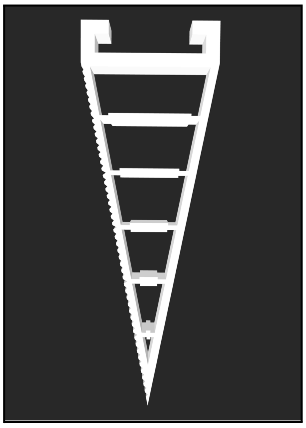

Fig10(a): Horizontal Ladder Fill with Microwedges

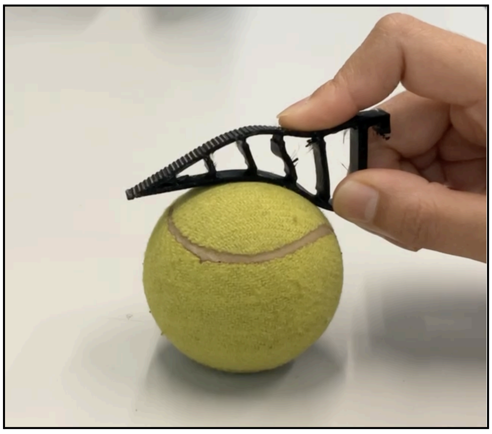

Fig10(b): Finger conforming around the ball

3.4 Outcomes, Results, and Decision-Making

3.4.1 What Worked

1. ***3-to-4 finger expansion:*** The additional finger improved grasping stability and distributed grip force more evenly across object surfaces.
2. ***Rack-and-pinion kinematics:*** Fully resolved the opening-range vs. closure trade-off by decoupling both parameters from pivot geometry.
3. ***Lateral guide walls:*** Eliminated tangential finger drift, restoring clean and symmetric actuation.
4. ***Structural reinforcements:*** Increased base thickness and side-wall support successfully resolved the fracture and separation failures in the pin-and-ray mechanism.
5. ***Part-splitting for SLA:*** Dividing the assembly into two printable sections accommodated the SLA geometric constraint without requiring a complete redesign.
6. ***Spike structures and internal fill geometry modifications:*** Spike structures improved grip friction on slippery surfaces.

3.4.2 What Did Not Work

1. ***Worm-follower pivot relocation:*** While it improved opening range, it introduced an irresolvable fingertip closure gap - the two objectives were geometrically coupled.
2. ***Curved-pivot and slit-angle corrections:*** Neither compensating approach fully restored closure without sacrificing opening range; the coupling was inherent to the mechanism.
3. ***ABS printing for the screw mechanism:*** Print was fragile and insufficient surface finish introduced excessive friction in the screw interface.

4. ***SLA as sole manufacturing solution:*** SLA lacked the structural durability to withstand repeated screw actuation pressure, resulting in fracture under operational loads.

3.4.3 Decision-Making Process

All design decisions were grounded in direct physical evidence from prototype tests rather than theoretical prediction. The key decision points were:

1. ***Worm-follower → Rack-and-Pinion:*** Taken after confirming that neither the curved-pivot nor the slit-angle approach could resolve the closure gap within the worm-follower architecture. The rack-and-pinion was selected because it eliminates the coupling at the kinematic level.
2. ***ABS → SLA:*** Driven by observed surface finish inadequacy in FDM prints causing friction in the screw mechanism.
3. ***Part-splitting:*** Chosen as the minimum-change fix to the SLA print constraint, preserving all finalised geometry.
4. ***SLA → Aluminium:*** Taken after confirming SLA fracture under screw actuation loads. Aluminium was identified as the only available material that simultaneously meets the surface finish and strength requirements.

4. Testing

To test our various gripper designs, we referred to the performance metrics derived for robotic arms and grippers by the National Institute of Standards and Technology.
For the evaluation against all of the metrics defined, a split cylinder is used. A split cylinder is a standardised testing equipment used for robotic grippers. It consists of two three-dimensional (3D) printed structures that can hold two or more single-axis force load cells (sensors) (Falco et al., 2020).

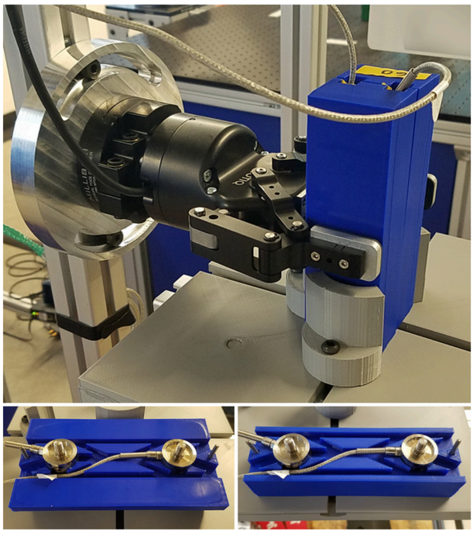

Fig 11: block split cylinder

Falco, J., Hemphill, D., Kimble, K., Messina, E., Norton, A., Ropelato, R., & Yanco, H. (2020). Benchmarking Protocols for Evaluating Grasp Strength, Grasp Cycle Time, Finger Strength, and Finger Repeatability of Robot End-effectors. IEEE Robotics and Automation Letters, 5(2). This author manuscript was made available in PubMed Central (PMC) on November 17, 2020

To measure the force exerted on the artifact, the load cells are oriented parallel to one another by the end-effector along the measurement axis (Falco et al., 2020). The design of the split cylinder is determined by the type of grasp. A block design is used for a pinch grasp, while the cylindrical design is used for wrap grasps (Falco et al., 2020). Due to our grippers exerting a wrap grasp, as detailed earlier in our literature review, we adapted the split cylinder to a cylinder design with a slot for the REES52 1kg wide bar Load Cell, as shown in Figure 12.

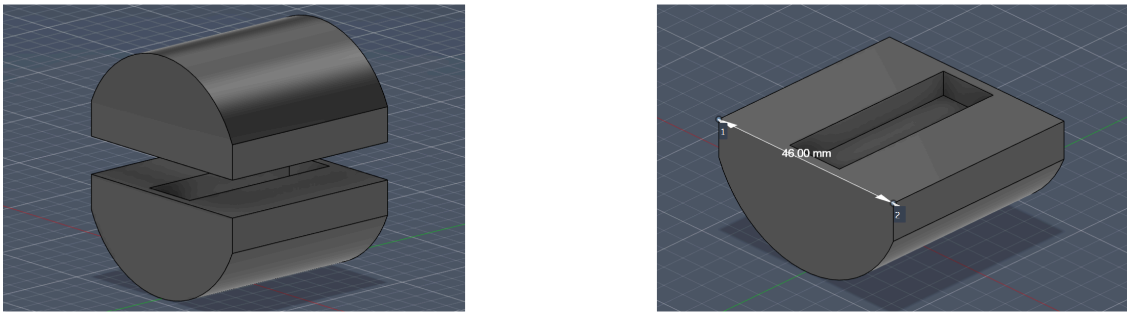

Fig 12: Our design of the split cylinder.

The results of the testing of all of our grippers will be presented in the end-of-semester presentation.

4.1 Testing Metrics

From the documentation by the National Institute of Standards and Technology, the metrics that are relevant and implementable for our project are grasp strength, slip resistance, and grasp cycle time. In this section, we focus on how we plan to proceed with the testing for these metrics.

4.1.1 Grasp strength:

Grasp Strength evaluates the maximum force a robotic hand can impose on an object, providing insight into payload capabilities. In order to evaluate it, a split cylinder is necessary.

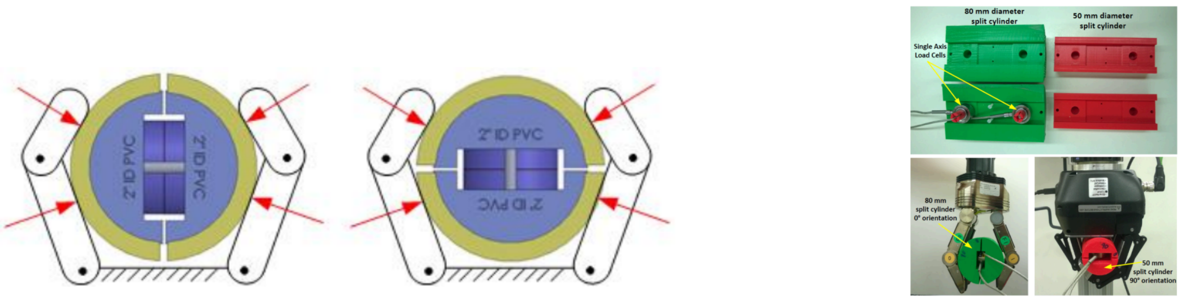

Fig 13: 13a: A diagrammatic representation of a split cylinder artifact in the 0° (left) and 90° (right) orientation, 13b: 3D printed 80mm and 50mm diameter prototypes of a split cylinder

(Falco, J., Van Wyk, K., & Messina, E. (2018). Performance metrics and test methods for robotic hands (DRAFT NIST Special Publication 1227). National Institute of Standards and Technology)

In this test, the gripper holds the maximum force it can exert for 5 seconds, then retracts to open fully. Repeat this process for 32 cycles for variously sized/oriented objects/artifacts. Additionally, ensure to ignore the first and last 10% of readings in 5 sec hold in order to not include quasistatic forces (Falco, Van Wyk, & Messina, 2018).

For each set of instantaneous force readings, we need to add forces across all load cells since they are in-line to yield a total grasp force Ftotal. This can be represented as:

$$F_{total} = \sum_{i=1}^{n} F_i$$

Because forces are distributed from multiple angles, this test is run twice in two different orientations: 0° (where the internal load cell axis is parallel to the hand's palm) and 90° (where the load cell axis is perpendicular to the palm) (Falco, Van Wyk, & Messina, 2018)

4.1.2 Slip resistance

Slip resistance tests a hand's ability to resist slipping by measuring the maximum pull force it can withstand under a full-force wrap grasp. In order to rest this, a cylindrical artifact is grasped at maximum power, and a controlled, increasing axial force is applied to the object until "gross slipping" is visually confirmed. The maximum pull force is taken note of. (Falco, Van Wyk, & Messina, 2018)

Materials required for this test are: Standard ASTM D2665 PVC pipe segments (or split cylinder artifacts), a single-axis load cell, a mechanism for applying a controlled increasing axial pull force (e.g., a linear drive coupled with a cable and in-line spring), and data acquisition systems (Falco, Van Wyk, & Messina, 2018)

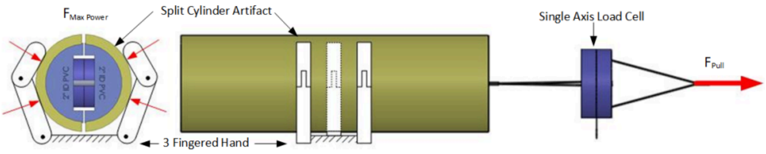

Fig: 14 A diagrammatic depiction of a three-finger wrap grasp on a cylindrical artifact under maximum power FMaxPower with the cylinder subjected to an axial pull force Fpull

(Falco, J., Van Wyk, K., & Messina, E. (2018). Performance metrics and test methods for robotic hands (DRAFT NIST Special Publication 1227). National Institute of Standards and Technology)

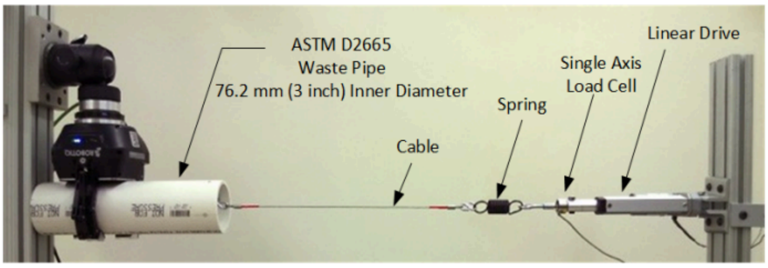

Fig 15 Prototype depicting slip resistance testing

(Falco, J., Van Wyk, K., & Messina, E. (2018). Performance metrics and test methods for robotic hands (DRAFT NIST Special Publication 1227). National Institute of Standards and Technology)

4.1.3 Grasp Cycle Time

The grasp cycle time measures the minimum time required for the gripper to achieve full closure from a "pre-grasp position" and return to the same position. For each test cycle, we record the pull force, Fpull, over time. From this force/time plot, we extract the maximum pull force Fpull max. This experiment is run on various pipe sizes, from which we calculate the mean and 95% confidence intervals for various diameter sizes. (Falco, Van Wyk, & Messina, 2018)

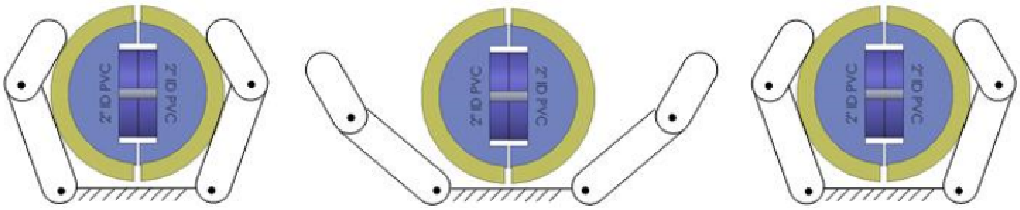

Fig 16 A diagrammatic representation of the split cylinder artifact in the dominant orientation relative to power grasp stages
(Falco, J., Van Wyk, K., & Messina, E. (2018). Performance metrics and test methods for robotic hands (DRAFT NIST Special Publication 1227). National Institute of Standards and Technology)

The materials needed for this test are a single-axis load cell for 1D force and a split cylinder artifact. As done in the grasp strength section, for each set of instantaneous force readings, add forces across all load cells to calculate force Ftotal. (Falco, Van Wyk, & Messina, 2018)

4.2 Objects to test

In order to carry out the tests, we will be using objects of various sizes, textures and shapes. A set of objects that we will be testing on includes a tennis ball, nails (small, medium, large), marbles, a wooden box, a foam brick, a Rubik's cube, a Pringles can and a mug, all of which can be found in the Yale-CMU-Berkeley Object Set. This is a widely established set providing physical objects and digital models for benchmarking. We will compare our results with submissions from similar grippers published (YCB Benchmarks, n.d.) Along with these objects, we will be testing on a wooden branch with algae to simulate slippery surfaces present underwater.

5. Conclusion

This project set out to design, prototype, and iteratively improve an underwater robotic gripper for an AUV capable of grasping a range of object types within the practical constraints of weight, water resistance, and single-motor actuation.

The fin-ray effect was selected as the core grasping principle because its passive mechanical compliance distributes contact force across a large area rather than concentrating it at a point, making it fundamentally better suited to fragile and irregular underwater specimens than rigid parallel-finger designs. This choice was validated by the literature and confirmed through physical testing: the TPU fin-ray fingers demonstrably conformed around curved objects, and the microwedge surface features on the ladder-fill variant improved grip friction on slippery surfaces, directly addressing the primary failure mode of underwater grasps identified by Mazzeo et al. (2022).

With regards to the design of the gripper, through our iterations, we initially explored a worm-follower architecture but ended up transitioning to a rack-and-pinion mechanism. This was due to the geometric coupling between the opening range and closure geometry in the worm-follower that could not be resolved through corrective modifications alone. Our transition to a rack-and-pinion mechanism eliminated this coupling at the kinematic level and delivered a robust design capable of simultaneous full closure and maximum opening range across all four fingers without dependency on pivot placement. Material selection was another aspect that proved equally critical and non-obvious. ABS failed at surface finish, and SLA failed at structural fatigue, causing us to finalise on a rack-and-pinion body in aluminium.

Our testing is currently underway as we plan on testing against three NIST SP 1227 metrics, grasp strength, slip resistance, and grasp cycle time, using a split cylinder artifact housing a REES52 1kg wide-bar load cell and interfaced with an HX711 amplifier.

Through the entire process of researching, prototyping, redesigning and testing, a significant outcome was not any single design decision, but the understanding of how engineering actually works. The circularity in this process of researching, prototyping, hitting a wall, stepping back, and redesigning from a new angle only to repeat it once again is not something that can be shortcut. Experiencing it firsthand across every layer of this project, from finger geometry to actuation mechanism to material selection to test infrastructure, is the most transferable thing this work produced.

Our work with the gripper is not yet complete. Post-testing, the primary direction that we see for future work is the adaptation of silicone projections onto the finger to further improve grip on the wet, smooth, and biologically irregular surfaces characteristic of the underwater operating environment. While this might not fall under the jurisdiction of ILGC-VI, we are hopeful that this direction and many others will be explored.

References:

Hawkes, E.W., Christensen, D.L., Han, A.K., Jiang, H. and Cutkosky, M.R., 2015. Grasping without squeezing: Shear adhesion gripper with fibrillar thin film. In *2015 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 2305–2312). IEEE.

Mazzeo, A., Aguzzi, J., Calisti, M., Canese, S., Angiolillo, M., Allcock, A.L., Vecchi, F., Stefanni, S. and Controzzi, M., 2022a. Marine robotics for deep-sea specimen collection: a taxonomy of underwater manipulative actions. *Sensors*, 22(4), 1471.

Mazzeo, A., Aguzzi, J., Calisti, M., Canese, S., Vecchi, F., Stefanni, S. and Controzzi, M., 2022b. Marine robotics for deep-sea specimen collection: a systematic review of underwater grippers. *Sensors*, 22(2), Article 648.

Srinivas, G.L., Javed, A. and Faller, L.M., 2024. Versatile 3D-printed fin-ray effect soft robotic fingers: lightweight optimization and performance analysis. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 46(6), p.382.

Falco, J., Van Wyk, K., & Messina, E. (October 2018). Performance Metrics and Test Methods for Robotic Hands (DRAFT NIST Special Publication 1227). National Institute of Standards and Technology, U.S. Department of Commerce

Falco, J., Hemphill, D., Kimble, K., Messina, E., Norton, A., Ropelato, R., & Yanco, H. (2020). Benchmarking Protocols for Evaluating Grasp Strength, Grasp Cycle Time, Finger Strength, and

Finger Repeatability of Robot End-effectors. IEEE Robotics and Automation Letters, 5(2). This author manuscript was made available in PubMed Central (PMC) on November 17, 2020.

CB Benchmarks. (n.d.). Results & records. YCB Benchmarks – Object and Model Set. https://www.ycbbenchmarks.com/world-records/